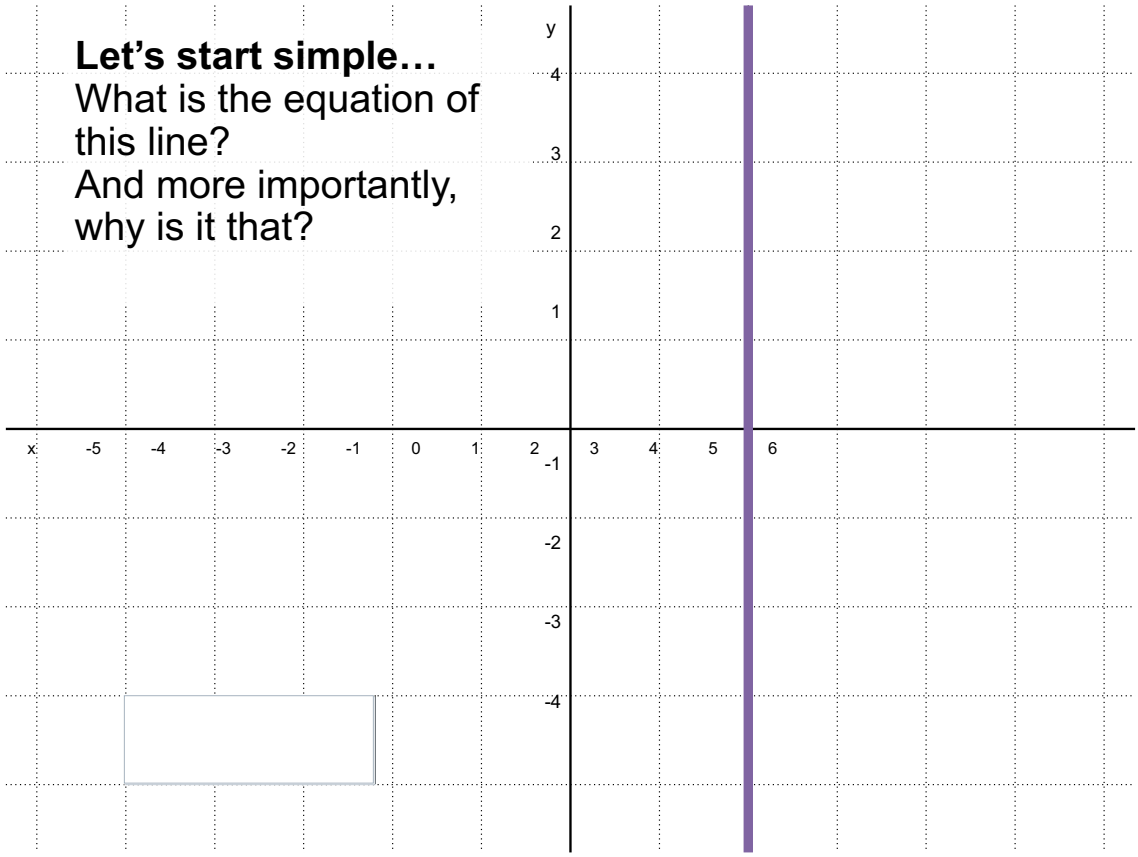
# 1.03 Coordinate Geometry: Linear Graphs

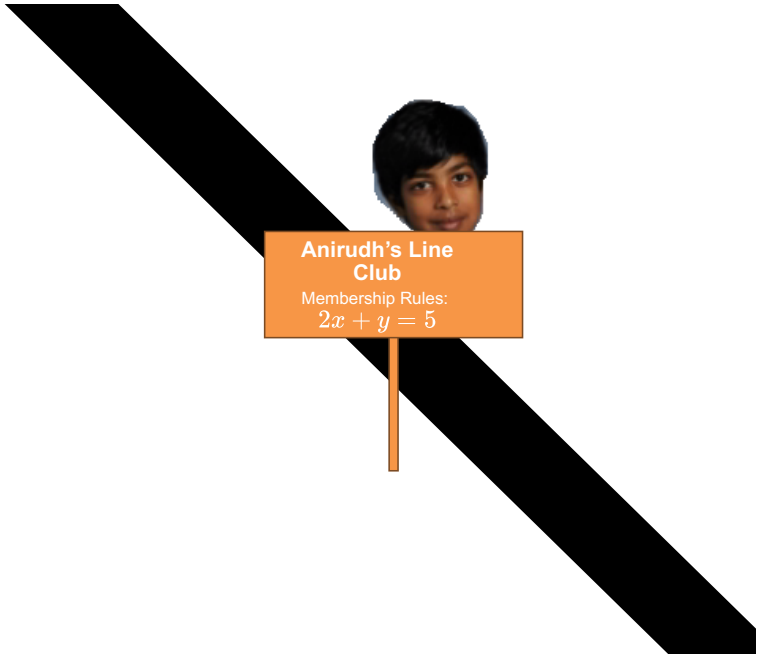
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Chapter 5 · Original slide 5

## Lines and Equations of Lines

! A line consists of all points which satisfy some equation in terms of $x$ and/or $y$.



# 1.03 Coordinate Geometry: Linear Graphs

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## Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of $a$.

Find the coordinate of the point where the line $2x + y = 5$ cuts the $x$-axis.

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## Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) $y$-axis:

b) $x$-axis:

What mistakes do you think it's easy to make?

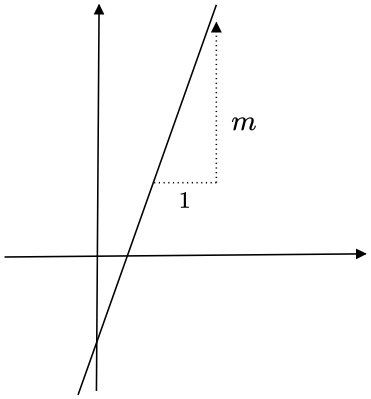
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## 1.03 Coordinate Geometry: Linear Graphs

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### Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what $y$ changes by as $x$ increases by 1.




So if the $y$ value increased by 6 as the $x$ value increased by 2, what is $y$ increasing by for each unit increase of $x$?
How would that give us a suitable formula for the gradient $m$?

$m =$ ☐

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### Examples

Find the gradient of the line that goes through the points:

1 $(1, 4)$ $(3, 10)$ $m =$ ☐

2 $(5, 7)$ $(8, 1)$ $m =$ ☐

3 $(2, 2)$ $(-1, 10)$ $m =$ ☐

4 Show that the points
$\mathrm{A}\,(3, 4), B\,(5, 5), C\,(11, 8)$ all lie on a straight

## 1.03 Coordinate Geometry: Linear Graphs

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### Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of $a$.

$\frac{a}{}$

$\frac{a}{}$

$a$

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### $y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

$y$-intercept

Why does it work?

# 1.03 Coordinate Geometry: Linear Graphs

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## $y = mx + c$

Determine the gradient and $y$-intercept of the line with equation $4x - 3y + 5 = 0$

$3y$

$y =$

Make $y$ the subject so we have the form $y = mx + c$
Put $y$ on the side it's positive.

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## $ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where $a, b, c$ are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where $a, b, c$ are integers.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$…

# 1.03 Coordinate Geometry: Linear Graphs

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## Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

### Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making $b$ zero.

$x + 4 = 0$

### Symmetry

In general, the '**linear combination**' of two variables $x$ and $y$ is $ax + by$, i.e. "some amount of $x$ and some amount of $y$". There is a greater elegance and symmetry to this form over $y = mx + c$ because $x$ and $y$ appear similarly within the expression.

### Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

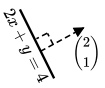
thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation $(1)$, the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation $(2)$, the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡ 


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## Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where $a, b, c$ are integers.

$5y$
$2x$

# 1.03 Coordinate Geometry: Linear Graphs

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## Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

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## Exercise 5B – Pure Y1 Page 92-93

8 A line $l$ cuts the $x$-axis at $(5, 0)$ and the $y$-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the $y$-axis at $-\frac{c}{b}$.

10 The line $l$ with gradient 3 and $y$-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

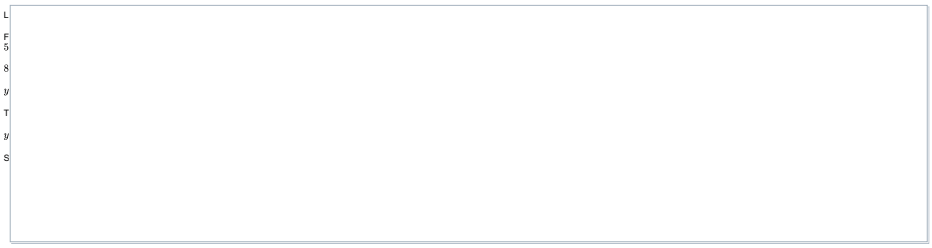
Find the values of $a$ and $c$. (2 marks)

## 1.03 Coordinate Geometry: Linear Graphs

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# Exercise 5B – Pure Y1 Page 92-93

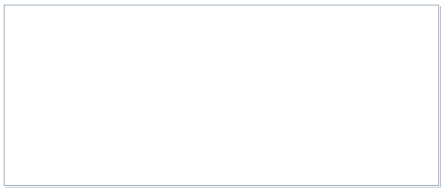
11 The straight line $l$ passes through $(0, 6)$ and has gradient $-2$. It intersects the line with equation $5x - 8y - 15 = 0$ at point $P$. Find the coordinates of $P$.

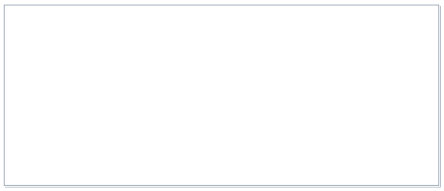


12 The straight line $l_1$ with equation $y = 3x - 7$ intersects the straight line $l_2$ with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of $b$.

b Find the value of $a$.



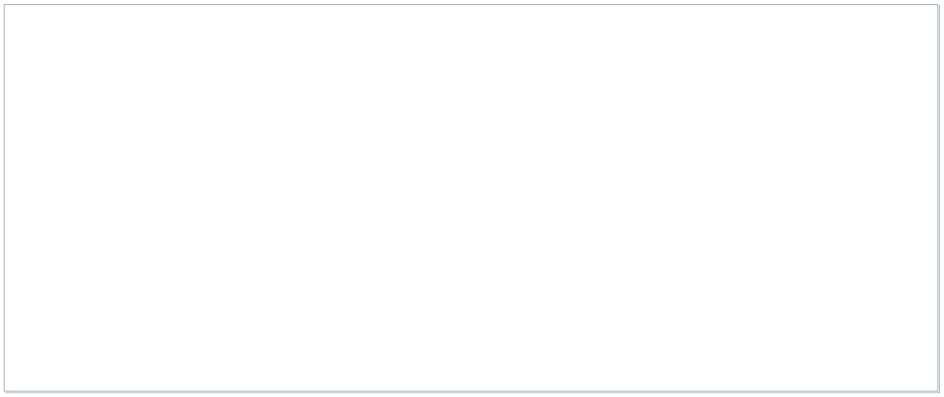


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# Equations using two points/point + gradient

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

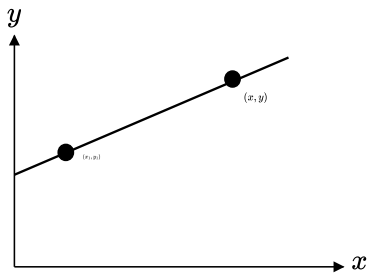
How would you have done this at **GCSE**?



# 1.03 Coordinate Geometry: Linear Graphs

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## A new way…




**Notes**: Note that $x_1$ and $y_1$ are constants while $x$ and $y$ are variables. The latter are variables because as these 'vary', we get different points on the line.

Suppose that $(x_1, y_1)$ is some fixed point on the line that we specify (e.g. $(3, 5)$).
Suppose that $(x, y)$ represents a generic point on the line, which is allowed to change as we consider different points on this line.
Then:

$m = \frac{y-y_1}{x-x_1}$

Thus:

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## Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

| Gradient | Point | (Unsimplified) Equation |
|---|---|---|
| $3$ | $(1, 2)$ | |
| $5$ | $(3, 0)$ | |
| $2$ | $(-3, 4)$ | |
| $\frac{1}{2}$ | $(1, -5)$ | |
| $9$ | $(-4, -4)$ | |

## 1.03 Coordinate Geometry: Linear Graphs

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# Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

**Test Your Understanding:**
Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

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# Exercise 5C – Pure Y1 Page 94-95

3 Find the equation of the line $l$ which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

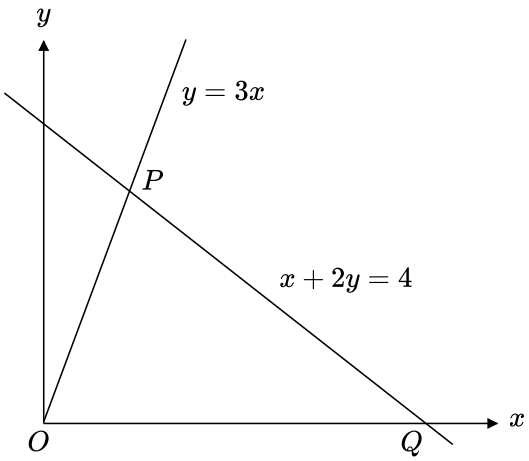
5 T $l$ $(a, 4)$ $(3a, 3)$. An equation of $l$ is $x + 6y + c = 0$. Find the value of $a$ and the value of $c$.

6 The straight line $l$ passes through $(7a, 5)$ and $(3a, 3)$. An equation of $l$ is $x + by - 12 = 0$. Find the value of $a$ and the value of $b$.

# 1.03 Coordinate Geometry: Linear Graphs

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## Intersection of lines




The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point $P$.

a) Determine the coordinates of $P$.

**We know from Chapter 4 we**

b

b

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## Test Your Understanding

C1 Edexcel May 2013 Q6

The straight line $L_1$ passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for $L_1$ in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.
\hfill(4)
The line $L_2$ has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of $L_1$ and $L_2$. \hfill(3)

$(-1, 3)$,
**(a)**
$m = \frac{y_2}{x_2}$
M1: Corr
$y - 3 =$
Correct s
$4y - 3x$
Or equiv
This A1 s

**(b)** Solve
Must rea
M1
$x = 3$ or
\textbf{B
**Fully cor**

# 1.03 Coordinate Geometry: Linear Graphs

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## Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the $y$-axis at the point $G$. Work out the coordinates of the point $G$.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point $T$. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point $T$. Write your answer in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.

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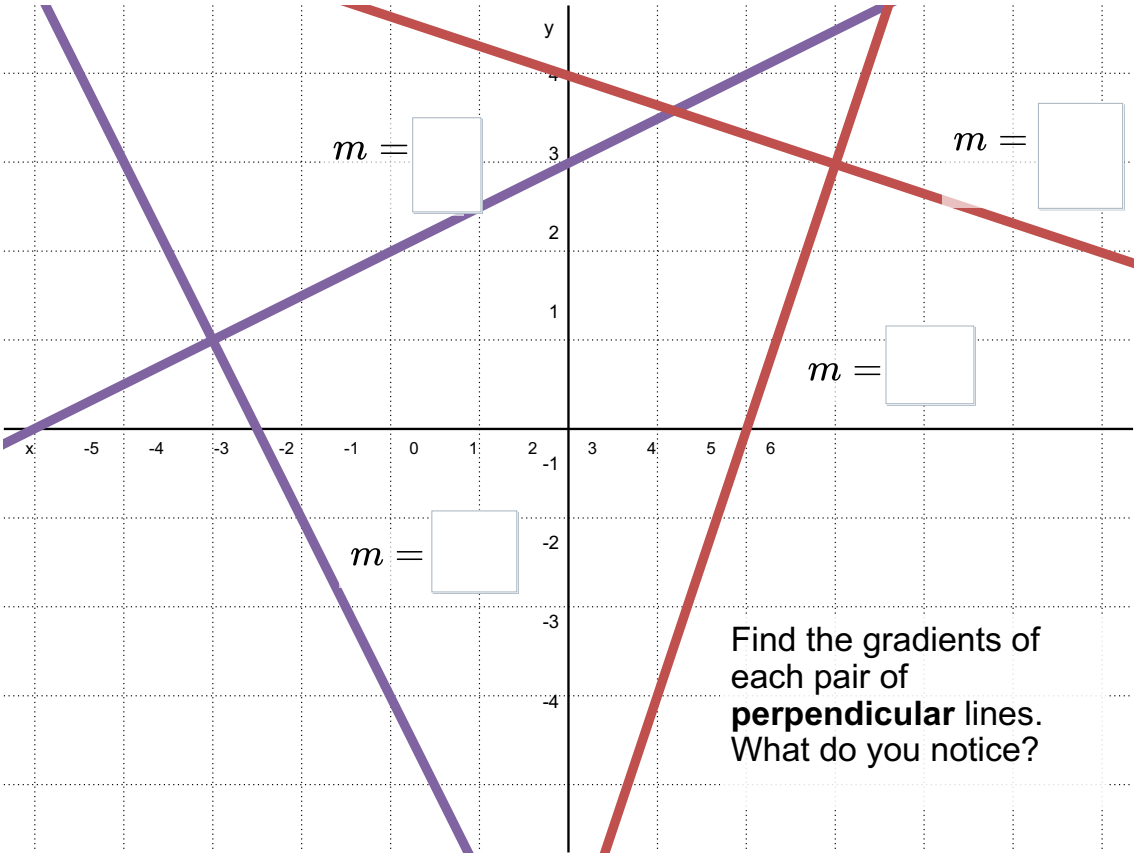
## Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the $y$-axis at the point $C$. The line $y = -3x - 15$ meets the $x$-axis at the point $D$. Find the equation of the line joining the points $C$ and $D$. Write your answer in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point $L$. The point $M$ has coordinates $(-3, 1)$. Find the equation of the line that passes through the points $L$ and $M$.

# 1.03 Coordinate Geometry: Linear Graphs

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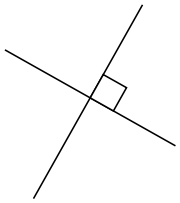
## Perpendicular Lines

The gradients of parallel lines are equal.
If two lines are perpendicular, then the gradient of one is the **negative reciprocal** of the other.

$$m_1 = -\frac{1}{m_2}$$

To **show** that two lines are perpendicular:

$$m_1 m_2 = -1$$



| Gradient | |
|---|---|
| $2$ | |
| $-3$ | |
| $\frac{1}{4}$ | |
| $5$ | |
| $-\frac{2}{7}$ | |
| $\frac{7}{5}$ | |

## 1.03 Coordinate Geometry: Linear Graphs

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# Example Problems

1 A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

2 A line $L_1$ goes through the [illegible] $A(1, 3)$ $B(3, -1)$. A second line $L_2$ is perpendicular to $L_1$ and passes through point B. Where does $L_2$ cross the x-axis?

$(\mathbf{5,0})$

3 Are the following lines para

$y = \frac{1}{2}x$

$2x - y + 4 = 0$

**Neither. Gradients are $\frac{1}{2}$ and 2. But $\frac{1}{2} \times 2 = 1$, not -1, so not perpendicular.**

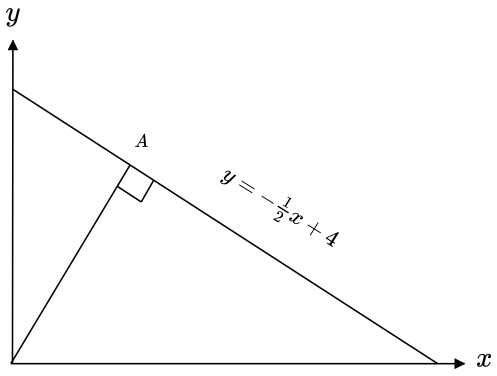
Chapter 5 · Original slide 31

# Test Your Understanding

1 A line goes through the point (4,7) and is perpendicular to another line with equation $y = 2x + 2$. What is the equation of the line? Put your answer in the form $ax + by + c = 0$, where $a, b, c$ are integers.

$\mathbf{y = -\frac{1}{2}x + 9}$

2 Determine the point $A$.



# 1.03 Coordinate Geometry: Linear Graphs

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## Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2), B(4, 8), C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

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## Exercise 5F – Pure Y1 Page 100

9 The line $l$ passes through the points $(-3, 0)$ and $(3, -2)$ and the line $n$ passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines $l$ and $n$ are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5), B(7, 1), C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

# 1.03 Coordinate Geometry: Linear Graphs

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## Exercise 5E/5F – Pure Y1 Page 96

### Extension Problems

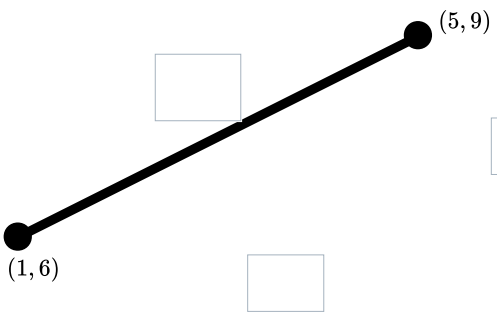
*[MAT 2004 1D]*
What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

*[MAT 2014 1D]* The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of $m$)

*[STEP I 2004 Q6]* The three points $A, B, C$ have coordinates $(p_1, q_1), (p_2, q_2)$ and $(p_3, q_3)$, respectively. Find the point of intersection of the line joining $A$ to the midpoint of $BC$, and the line joining $B$ to the midpoint of $AC$. Verify that this point lies on the line joining $C$ to the midpoint of $AB$.
The point $H$ has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line $AH$ intersects the line $BC$ at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line $BH$ intersects the line $AC$ at right angles.
Deduce that if $AH$ is perpendicular to $BC$ and also $BH$ is perpendicular to $AC$, then $CH$ is perpendicular to $AB$.

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## Distances between points




How could we find the **distance** between these two points?
Hint:

# 1.03 Coordinate Geometry: Linear Graphs

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## Examples

**Distance between:**

$(3, 4)$ and $(5, 7)$
$(5, 1)$ and $(6, -3)$
$(0, -2)$ and $(-1, 3)$

**Note**: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

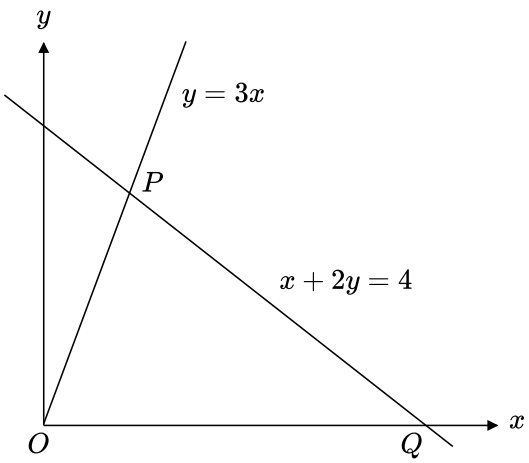
Quickfire Questions:

**Distance between:**

$(1, 10)$ and $(4, 14)$
$(3, -1)$ and $(0, 1)$
$(-4, -2)$ and $(-12, 4)$

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## Area of Shapes




The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point $P$.

a) Determine the coordinates of $P$.

*(We did this in a previous lesson)*
**Just solve two equations simultaneously.**
$\mathbf{x+2\,(3x)=4}$
$\mathbf{7x=4}$
$\mathbf{x=\frac{4}{7}}$
$\mathbf{y=3\,\left(\frac{4}{7}\right)=\frac{12}{7}}$

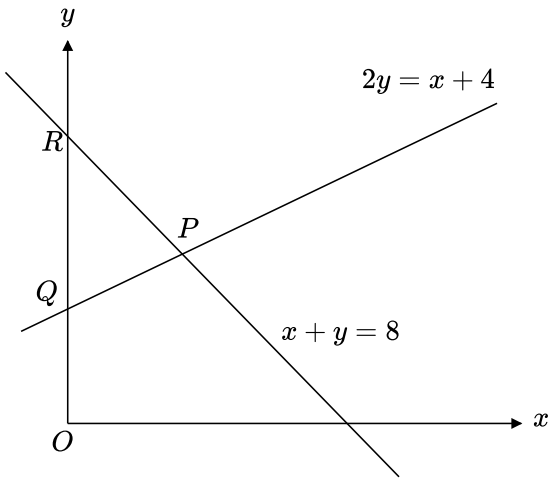
b) The line $x + 2y = 4$ intersects the $x$-axis at the point $Q$. Determine the area of the triangle $OPQ$.
**When $\mathbf{y=0, x=4}$**

## 1.03 Coordinate Geometry: Linear Graphs

$$\text{Area} = \frac{1}{2} \times 4 \times \frac{12}{7} = \frac{24}{7}$$

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# Further Example



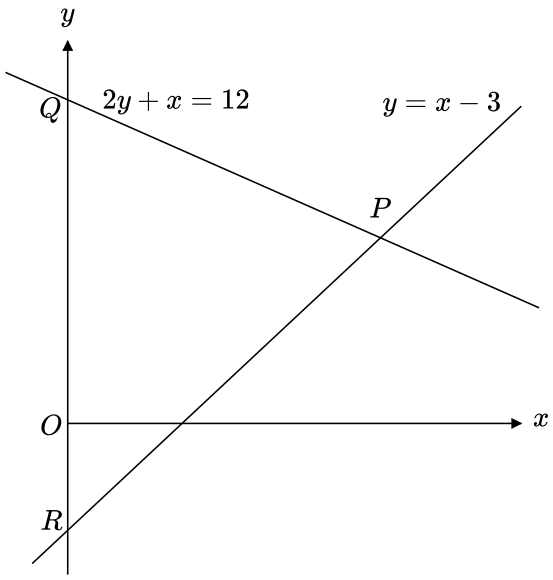

**Tip**: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

a Determine the length of $PQ$.

b Determine the area $PQR$.

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# Test Your Understanding




a) Determine the coordinate of $P$.

$\mathbf{2}$ (illegible)

$\mathbf{x}=$

b) Determine the area of $PQR$.

$\mathbf{R}\,(0,-3)$

$\mathbf{Q}$

$\mathbf{A}$

c)

$\mathbf{P}\,(6,3), \quad \mathbf{Q}\,(0,6)$

$\mathbf{Dist} = \sqrt{6^2+3^2} = \sqrt{45}$

# 1.03 Coordinate Geometry: Linear Graphs

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## Exercise 5G – Pure Y1 Page 102

7 A point $P$ lies on the line with equation $y = 4 - 3x$. The point $P$ is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point $P$. \hfill(5 ma...

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle $ABC$.

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## Exercise 5G – Pure Y1 Page 103

12 The straight line $l_1$ passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for $l_1$ in the form $ax + by + c = 0$, where $a, b$ and $c$ are integers.
b Write down the coordinates of $A$, the point where straight line $l_1$ crosses the $y$-axis.
The straight line $l_2$ passes through the origin and has gradient 3. The lines $l_1$ and $l_2$ intersect at the point $B$.
c Calculate the coordinates of $B$.
d Calculate the exact area of $\triangle OAB$.

## 1.03 Coordinate Geometry: Linear Graphs

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# Exercise 5G – Pure Y1 Page 102-103

**Extension Problems**

*[MAT 2001 1C]*
The shortest distance from the origin to the line $3x + 4y = 25$ is what?